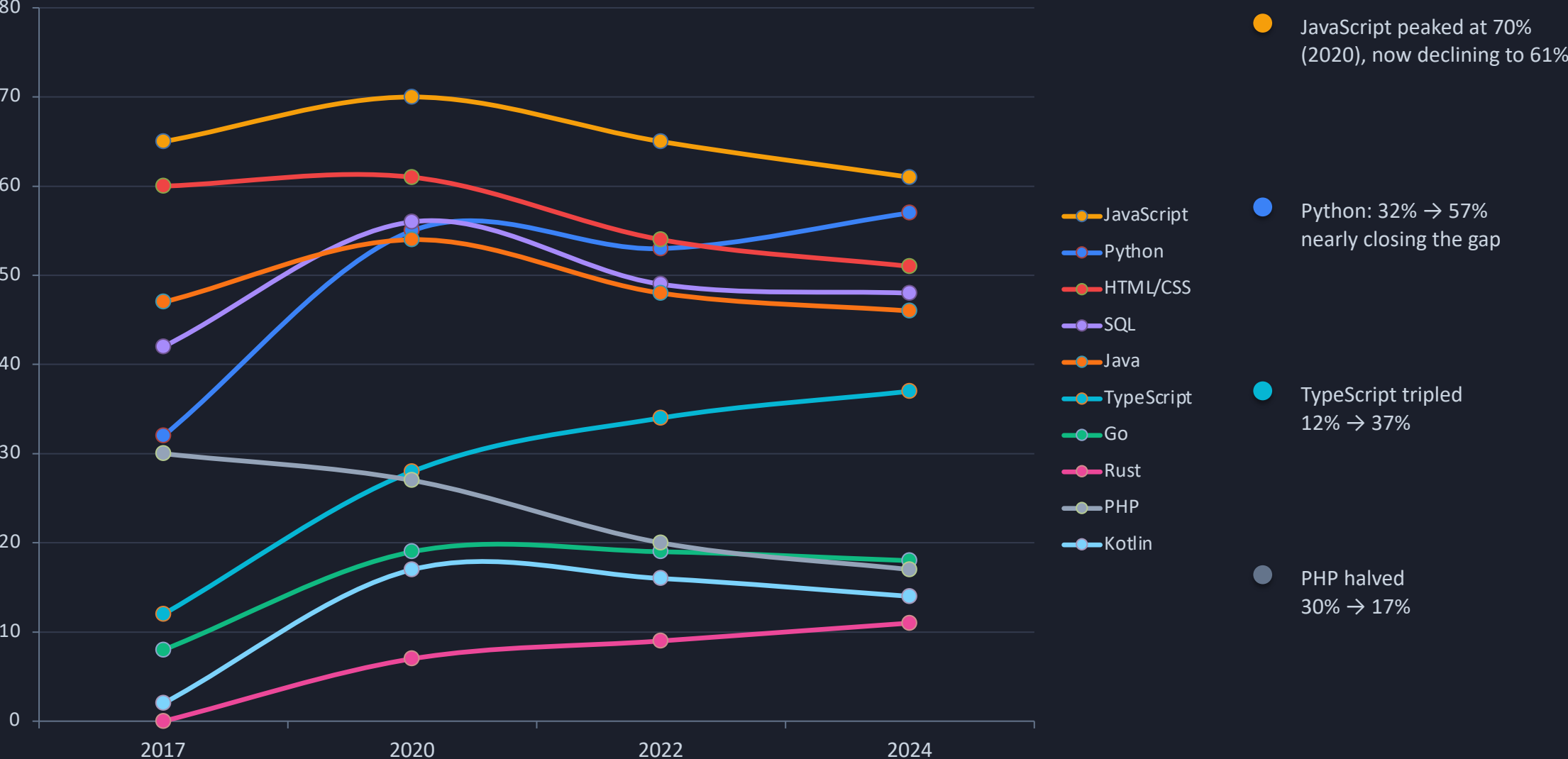

State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

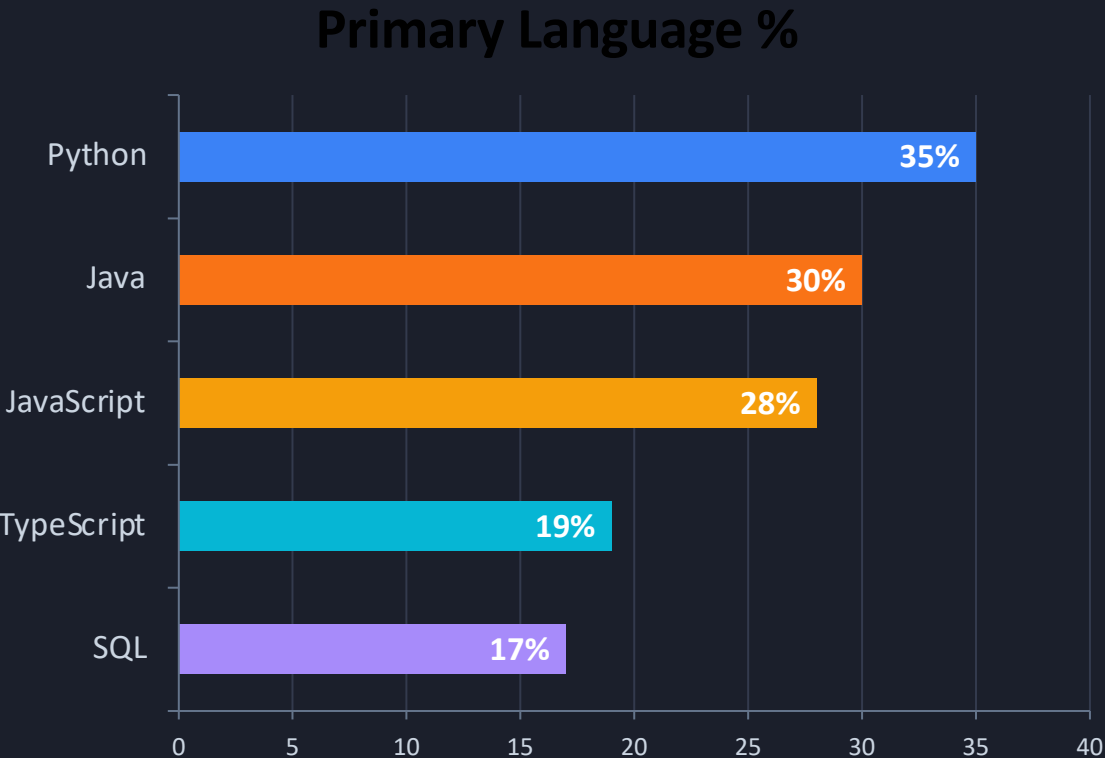
Based on JetBrains' annual survey of 23,262 developers worldwide

Python Closes the Gap: 8 Years of Language Usage Shifts

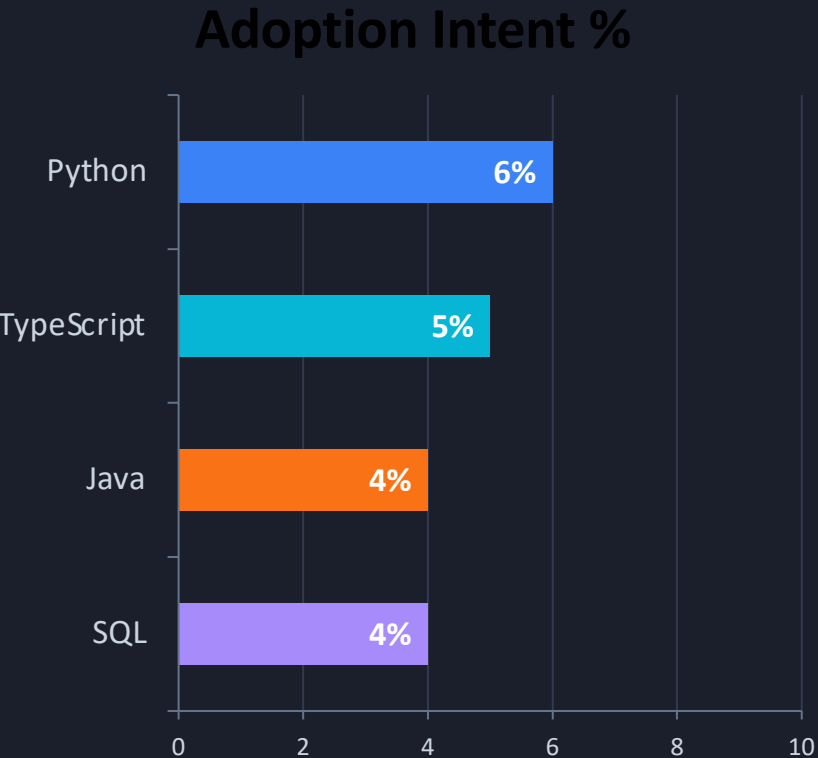


Python Is Now the #1 Primary Language at 35%

Primary Language Choice

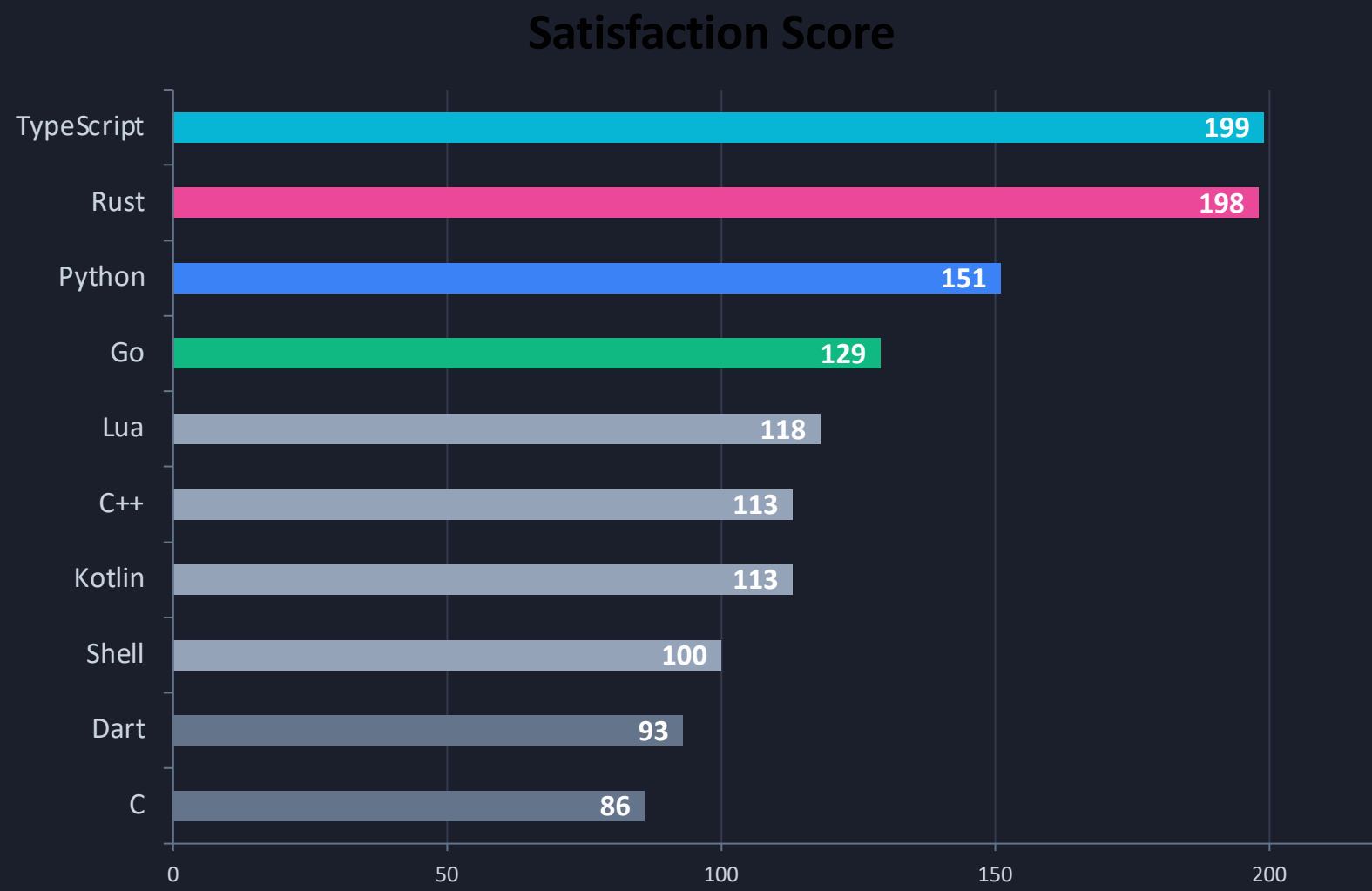


Adoption Intent (Planning to Adopt)



Developers stick with what they know — adoption intent for new primary languages stays below 6% across the board. Ecosystem lock-in and expertise investment keep developers loyal to their chosen stacks.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



KEY INSIGHT

TypeScript (199) and Rust (198) score nearly identically as the most loved languages.

Both are prized for type safety, modern tooling, and expressive type systems.

Satisfaction correlates with type safety and developer experience quality.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Top Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

DOMAIN INSIGHTS

Python dominates AI/ML at 31% and data science at 28% — its rise correlates directly with the AI boom.

TypeScript (62%) and JavaScript (58%) together own user interface development.

Java remains the backbone for APIs & services (49%) and application logic (55%).

C++ leads system tools (27%), hardware interfacing (21%), and graphics (15%).

Language choice is driven by domain, not general preference.

What This Means for Your Technology Strategy in 2025

1

Invest in Python for AI/ML Teams

Python is the undisputed leader — 31% AI/ML share, 28% data science. The AI boom makes Python expertise non-negotiable.

2

TypeScript Is the Future of Frontend & Full-Stack

TypeScript tripled usage to 37% and leads developer satisfaction at 199. It's the clear successor for JavaScript-heavy teams.

3

Rust and Go: Growth Plays for Infrastructure

Both designed for performance, concurrency, and compiler safety. Rust rising to 11%, Go stable at 18% — target for systems and cloud-native hiring.

4

PHP and Ruby Hiring Pools Are Shrinking

PHP dropped from 30% to 17%; Ruby from 10% to 4%. Plan succession strategies for legacy codebases in these ecosystems.

5

Satisfaction Drives Retention

Developer satisfaction correlates with type safety and modern tooling. Factor this into technology choices — it directly impacts team retention.

Source: JetBrains State of Developer Ecosystem 2024 | 23,262 respondents | jetbrains.com/lp/devecosystem-2024